

Cyclistic User Behavior Analysis

Converting Casual Riders to Annual Members Through Data-Driven
Marketing Strategy

The Business Task

The Core Challenge

How do annual members and casual riders use Cyclistic bikes differently, and how can we use that data to convert casuals into members?

High-Profit Potential: Cyclistic's current financial models reveal that annual members are significantly more profitable than single-ride or full-day pass users.

Targeted Conversion: Rather than targeting entirely new audiences, our goal is to design a targeted marketing strategy aimed directly at converting the existing casual rider base.



Data Integrity & Foundation

3.9M

Cleaned Trips

12 Months of Historical Data

To ensure high data integrity, the raw 12-month dataset was rigorously processed using SQL before visualization.

- ✓ Consolidated 12 separate monthly tables.
- ✓ Removed records with missing critical station data.
- ✓ Filtered out invalid trips (false starts under 1 minute and unreturned bikes over 24 hours).

Result: A highly accurate, executive-ready dataset.

The Discovery: Commuters vs. Cruisers

The Weekday Commuter

Annual Members

-  **High Volume:** 2.53 Million total rides.
-  **Short Duration:** Averages ~12.5 minutes per ride.
-  **Mid-Week Peak:** Usage remains consistently high Monday through Friday.
-  **Commute Times:** Peaks at 8:00 AM and 5:00 PM.

The Weekend Cruiser

Casual Riders

-  **Lower Volume:** 1.36 Million total rides.
-  **Long Duration:** Averages ~21.8 minutes (nearly double members).
-  **Weekend Surge:** Massive spike strictly on Saturdays (288k+) and Sundays.
-  **Leisure Times:** Steadily climbs throughout the mid-afternoon.

Visualizing The Weekend Surge

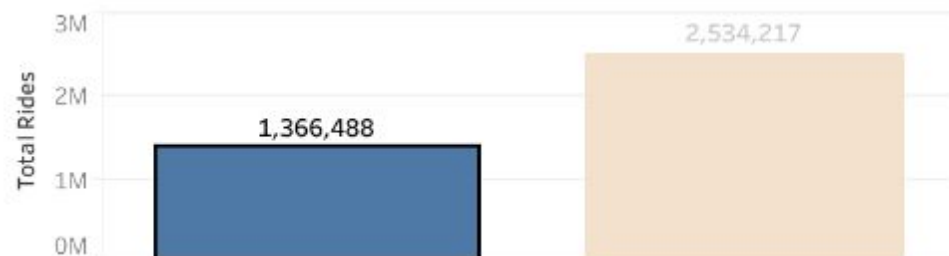
Cyclistic User Behavior: The Weekday Commuter vs. The Weekend Cruiser

An analysis of 3.9 million rides to inform casual-to-member conversion strategies

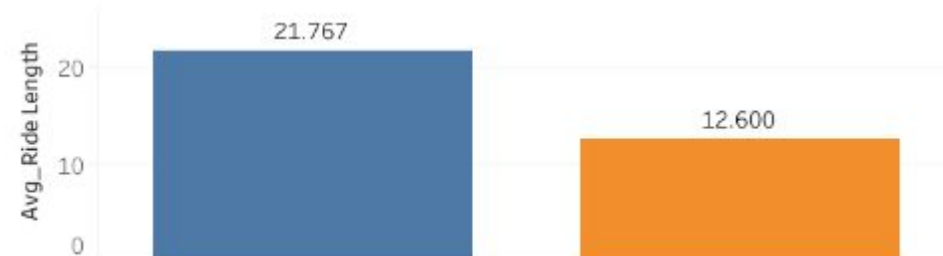
User Type

All

Total Ride Volume by User Type



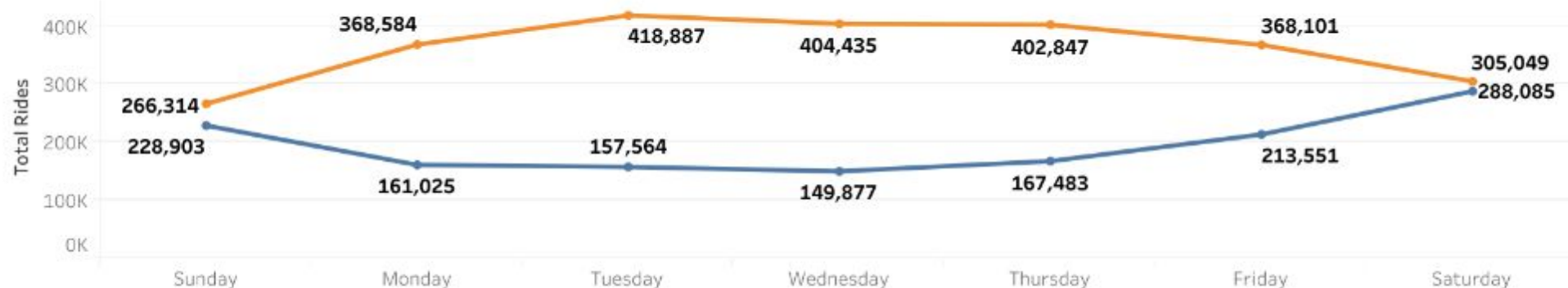
Average Ride Duration (Minutes)



Legends

casual
member

Weekly Usage Trends by Day



Strategic Recommendations



"Weekend Cruiser" Tier

Casual riders dominate the weekends but avoid weekday commuting. Introduce a Friday-Sunday membership tier to bridge the gap between single-ride passes and full commuter memberships.



Emphasize Duration Savings

Casual riders spend roughly 22 minutes per ride. Marketing campaigns should mathematically prove to them how an annual membership eliminates high per-minute surcharges on long rides.



Post-Ride Upgrade Prompts

Trigger targeted in-app notifications. When a casual user completes a trip exceeding 20 minutes on a weekend, instantly notify them of the exact cash savings an annual membership provides.

Measuring Success (KPIs)

To track the effectiveness of these marketing strategies, the data analytics team will monitor the following Key Performance Indicators post-launch:

Casual-to-Member Conversion Rate

Track the overall trip volume ratio month-over-month to verify if the current 65/35 (Member/Casual) split shifts toward members.

Average Ride Length of New Members

Monitor the duration of newly converted members. If it stays high (~21 mins), it validates we captured the "leisure" audience.

Weekend Membership Growth (YoY)

Measure the year-over-year increase in member rides strictly on Saturdays and Sundays to validate the weekend-targeted tier.



Questions?

Thank you for your attention.